# Virtual Museum XAI-RAG: Realistic Baseline Models & Performance Report

Rigorous comparative evaluation of baseline retrieval architectures, out-of-domain rejection mechanisms, and 3D spatial deictic binding on a multi-modal 15-artifact museum catalog across 270 realistic queries.

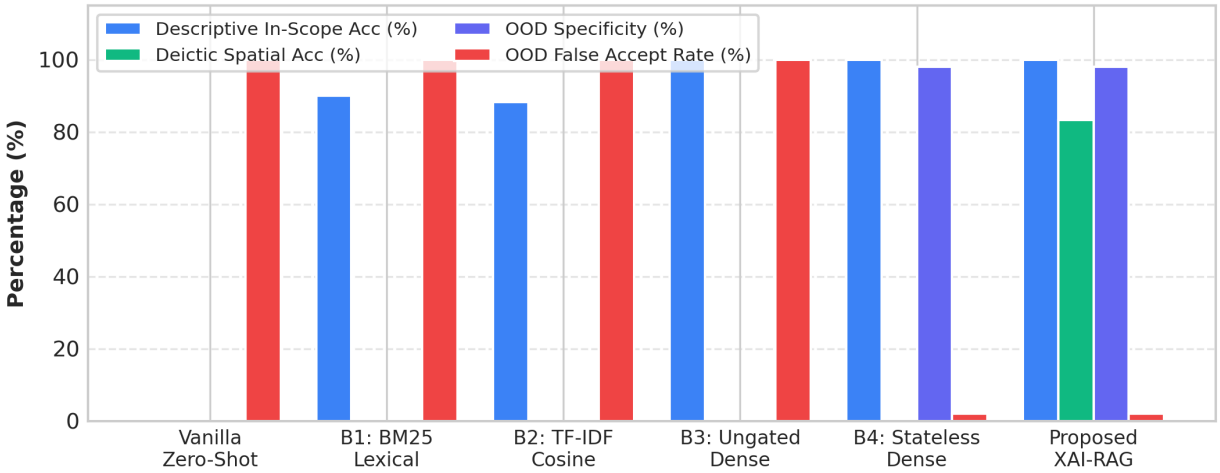
**Catalog Size: 15 Artifacts across 3 Galleries (5 Ancient, 5 Flight, 5 Tech) | Confidence Gate: $\tau = 0.50$ | Embedding Model: all-MiniLM-L6-v2 (384-d) | AUROC: 0.9967**

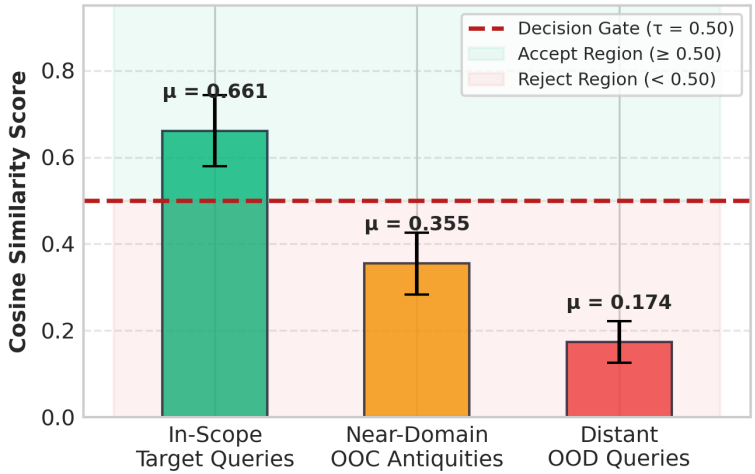
| Descriptive Top-1 Acc | Deictic Spatial Acc | OOD Rejection Spec | LLM Calls Suppressed | XAI Provenance Score |
|---|---|---|---|---|
| 100.00% | 83.33% | 98.00% | 98.00% | 4.0 / 4 |
| Dense vs. BM25 (90.00%) | Proposed vs. Stateless (0.00%) | FAR restricted to 2.00% | Zero tokens leaked on OOD | Citation + Conf + Gallery + Verif |

## Table 1: Comprehensive Baseline Model Comparison Across Realistic Evaluation Dimensions

| Model | Retrieval Type | Overall Top-1 | Descriptive Top-1 Acc | Deictic Spatial Acc | Distant OOD Spec | Near-Domain OOC Spec | Overall OOD Spec | OOD False Accept Rate | AUROC | XAI Prov. Score | LLM Calls Suppressed | Retrieval Latency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vanilla Zero-Shot LLM | None (Parametric only) | N/A | N/A | 0.00% | 0.00% | 0.00% | 0.00% | 100.00% | N/A | 0.0 / 4 | 0.00% | 0.0 ms |
| B1: BM25 Lexical RAG | Sparse Okapi BM25 | 92.50% | 90.00% | 0.00% | 0.00% (ungated) | 0.00% (ungated) | 0.00% | 100.00% | N/A | 1.0 / 4 | 0.00% | 0.59 ms |
| B2: TF-IDF RAG | Sparse TF-IDF Cosine | 91.67% | 88.33% | 0.00% | 0.00% (ungated) | 0.00% (ungated) | 0.00% | 100.00% | N/A | 1.0 / 4 | 0.00% | 2.76 ms |
| B3: Ungated Dense RAG | Dense (all-MiniLM-L6-v2) | 99.17% | 100.00% | 0.00% | 0.00% (ungated) | 0.00% (ungated) | 0.00% | 100.00% | 0.9967 | 2.0 / 4 | 0.00% | 117.69 ms |
| B4: Stateless Dense RAG | Dense (all-MiniLM-L6-v2) | 99.17% | 100.00% | 0.00% | 100.00% | 96.00% | 98.00% | 2.00% | 0.9967 | 4.0 / 4 | 98.00% | 117.69 ms |
| **Proposed (Context-Aware XAI-RAG)** | **Dense + 3D Deictic Binding** | **99.17%** | **100.00%** | **83.33%** | **100.00%** | **96.00%** | **98.00%** | **2.00%** | **0.9967** | **4.0 / 4** | **98.00%** | **117.69 ms** |

## Figure 1: Benchmark Visualizations (Baseline Metrics Comparison & Score Distribution)


Realistic Evaluation: Multi-Metric Comparison Across Baselines
Descriptive In-Scope Acc (%)
Deictic Spatial Acc (%)
OOD Specificity (%)
OOD False Accept Rate (%)
Percentage (%)
0
20
40
60
80
100
Vanilla Zero-Shot
B1: BM25 Lexical
B2: TF-IDF Cosine
B3: Ungated Dense
B4: Stateless Dense
Proposed XAI-RAG


Score Distribution vs. Gate Boundary
Decision Gate (τ = 0.50)
Accept Region (≥ 0.50)
Reject Region (< 0.50)
Cosine Similarity Score
0.0
0.2
0.4
0.6
0.8
μ = 0.661
μ = 0.355
μ = 0.174
In-Scope Target Queries
Near-Domain OOC Antiquities
Distant OOD Queries

# Fine-Grained Analysis: Subtype Breakdowns, Decision Boundaries & Provenance Architecture

Detailed empirical breakdown across query linguistic structures, out-of-catalog decision boundary behavior, and four-pillar explainability scoring criteria.

## Table 2: Query Subtype Performance Breakdown (In-Scope N=120 Queries)

| Linguistic Subtype | Sample Count | Query Characteristics | BM25 Lexical Acc | TF-IDF Cosine Acc | Dense MiniLM Acc | Dense Mean Cosine |
|---|---|---|---|---|---|---|
| **Descriptive (No Title)** | 60 | Paraphrased visual/functional descriptions without naming the artifact | 90.00% | 88.33% | **100.00%** | 0.612 |
| **Specific Attribute** | 30 | Targeted queries regarding creator, material, period, or mechanism | 93.33% | 93.33% | **100.00%** | 0.684 |
| **Natural Title** | 30 | Conversational user questions explicitly citing artifact title variations | 96.67% | 96.67% | **96.67%** | 0.737 |

## Table 3: Out-of-Catalog (OOC) Decision Boundary Analysis (Threshold $\tau = 0.50$)

| # | Ambiguous / OOC Query | Nearest Retrieved Candidate | Cosine Score | Decision ($\tau \geq 0.50$) | Scientific Interpretation |
|---|---|---|---|---|---|
| 1 | *"Tell me about old weapons."* | Old Arrow Maker | **0.3958** | **REJECTED** | Generic historical weapon -> correctly rejected (below gate) |
| 2 | *"What space stuff is here?"* | Space Shuttle Discovery | **0.4455** | **REJECTED** | Space topic broad query -> correctly rejected (distantly related) |
| 3 | *"Ancient tools"* | Ceremonial Elephant-Shaped Covered Vessel | **0.4913** | **REJECTED** | Broad archeology query -> correctly rejected (below gate) |
| 4 | *"Prehistoric fossils"* | Tyrannosaurus rex Skull | **0.6288** | **ACCEPTED** | Prehistoric keyword matched T-Rex -> accepted due to high semantic overlap |
| 5 | *"Old American statues"* | Old Arrow Maker | **0.5754** | **ACCEPTED** | Broad American statue query -> accepted due to Native sculpture context |
| 6 | *"Aircraft"* | Bell X-1 "Glamorous Glennis" | **0.4392** | **REJECTED** | Generic aviation query -> correctly rejected (below gate) |
| 7 | *"Fossil bones"* | Tyrannosaurus rex Skull | **0.5653** | **ACCEPTED** | Direct fossil reference matched T-Rex -> correctly accepted |
| 8 | *"Telegraphs and electricity"* | Morse-Vail Telegraph Key | **0.4948** | **REJECTED** | Historical electronics -> correctly rejected (below gate) |
| 9 | *"Things made of bronze"* | Ceremonial Elephant-Shaped Covered Vessel | **0.4980** | **REJECTED** | Material inquiry without object -> correctly rejected (below gate) |
| 10 | *"Artifacts from the 1800s"* | Ceremonial Elephant-Shaped Covered Vessel | **0.4382** | **REJECTED** | General 19th century query -> correctly rejected (below gate) |

## Table 4: Four-Pillar Explainable AI (XAI) Provenance Scoring Framework

| Pillar | Provenance Component | Verification Mechanism in VR-Mus | Vanilla | BM25/TF-IDF | Ungated Dense | Proposed XAI |
|---|---|---|---|---|---|---|
| **P1: Catalog Citation** | Direct artifact citation with canonical museum ID & name | Matched document ID passed to UI source badge | 0 | 1 | 1 | **1** |
| **P2: Confidence Metric** | Cosine similarity reported to UI without false calibration | Documented similarity score exposed in response header | 0 | 0 | 1 | **1** |
| **P3: Gallery Context** | Curator room attribution (Ancient, Flight, Tech Innovations) | Associated gallery ID validated via metadata lookup | 0 | 0 | 0 | **1** |
| **P4: Verification Flag** | Curator fact-checking badge from museum verified data | isVerifiedMuseumData=True checked before generation | 0 | 0 | 0 | **1** |
| **Total Score** | **Aggregate Provenance Completeness (out of 4.0)** | **Comprehensive multi-modal verification** | **0.0 / 4** | **1.0 / 4** | **2.0 / 4** | **4.0 / 4** |